

LONDRINA, Brazil

WMO: 837680

Lat: 23.334S

Lon: 51.130W

Elev: 1867

StdP: 13.73

Time Zone: -3.00 (SAB)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
6	46.7	50.1	35.0	31.9	60.7	39.2	37.8	64.4	15.1	69.8	13.4	69.3	5.0	220	0.290

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	15.9	93.6	71.0	91.7	71.4	89.8	71.7	78.1	84.5	77.2	83.8	76.4	83.0	8.0	30

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
76.9	150.4	80.8	75.5	143.1	79.5	74.9	140.4	79.1	43.2	85.1	42.2	83.3	41.4	83.5	84.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 15.3	13.3	11.7	40.8	97.5	3.9	2.3	38.0	99.1	35.7	100.5	33.5	101.7	30.6	103.4		
(5)			37.8	80.3	4.1	1.7	34.8	81.5	32.3	82.6	30.0	83.5	27.0	84.8		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	72.8	77.3	77.8	77.1	74.0	67.7	65.5	65.6	69.0	71.9	74.7	75.7	77.4
	DBStd	6.52	3.07	3.26	3.36	4.58	5.00	5.38	5.55	5.83	6.28	5.20	3.97	3.59
	HDD50	1	0	0	0	0	0	0	1	0	0	0	0	0
	HDD65	196	0	0	0	3	30	58	60	31	13	1	0	0
	CDD50	8316	848	778	840	719	549	466	486	590	656	765	771	848
	CDD65	3035	383	358	375	271	115	73	80	156	219	301	321	383
	CDH74	23823	2779	2705	2854	1999	650	390	551	1383	2097	2622	2643	3150
	CDH80	8343	919	967	1015	639	120	39	91	441	892	1082	977	1161
(14) Wind	WSAvg	5.1	5.4	4.9	4.9	4.6	4.4	4.1	4.5	5.1	5.8	6.1	6.2	5.6
(15) Precipitation	PrecAvg	60.3	9.1	7.6	5.0	3.6	4.6	3.4	2.6	1.8	4.8	5.1	5.9	6.8
	PrecMax	84.0	16.8	14.4	12.1	8.6	12.8	14.0	9.3	7.2	13.1	13.1	14.8	12.0
	PrecMin	47.3	1.9	2.4	0.8	0.0	0.6	0.0	0.0	0.6	0.2	2.0	0.5	0.5
	PrecStd	9.3	3.7	3.1	2.6	2.0	3.1	3.1	2.4	1.8	2.8	3.3	2.9	3.2
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	93.2	95.1	93.2	91.7	86.3	83.9	86.0	91.2	96.7	97.2	94.8	93.5
		MCWB	74.4	71.1	73.9	72.0	69.1	68.6	65.0	65.5	68.2	70.7	71.5	72.6
	2%	DB	89.6	91.1	90.0	89.2	82.8	80.8	82.5	87.7	93.0	93.4	91.3	91.3
		MCWB	74.7	73.5	73.6	71.8	69.4	68.0	64.9	65.1	66.8	71.2	71.8	73.0
	5%	DB	87.5	88.1	88.0	86.2	80.5	78.6	80.1	84.6	89.4	89.7	88.1	89.2
		MCWB	74.5	74.4	73.2	71.1	68.4	67.1	64.9	64.8	66.6	70.7	70.7	73.2
	10%	DB	85.1	86.0	85.9	84.0	77.3	75.6	77.0	82.2	85.6	86.0	85.9	86.2
		MCWB	74.2	74.2	72.8	70.7	67.1	66.2	64.4	64.4	66.0	70.0	70.7	72.7
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	79.3	79.3	79.2	76.8	73.4	72.8	70.9	72.2	74.0	76.5	77.6	78.4
		MCDB	84.5	84.6	85.3	83.1	81.4	78.0	77.5	80.1	84.5	85.4	85.5	85.8
	2%	WB	77.8	77.9	77.2	75.1	71.7	70.2	68.9	69.3	72.0	74.9	76.1	77.1
		MCDB	83.7	83.7	84.0	82.0	78.6	76.5	76.1	78.5	82.0	84.9	83.6	84.8
	5%	WB	76.8	76.8	76.1	73.7	70.2	68.8	67.4	67.6	70.5	73.6	74.7	76.1
		MCDB	82.9	82.6	82.8	81.3	77.0	75.2	74.9	78.1	80.0	83.2	82.0	83.3
	10%	WB	75.8	75.8	75.0	72.6	68.6	67.5	65.9	66.1	68.9	72.4	73.5	75.0
		MCDB	81.7	81.6	81.5	80.1	75.2	73.7	73.1	77.0	78.3	81.5	80.4	81.8
(35) Mean Daily Temperature Range	5% DB	MDBR	15.4	15.9	17.3	18.1	17.2	17.4	19.6	22.1	21.3	19.9	19.1	17.0
		MCDBR	18.9	19.4	20.2	20.6	20.6	20.7	23.6	26.0	26.9	25.3	22.9	20.9
		MCWBR	7.8	7.2	7.6	7.5	9.1	9.3	9.9	9.3	8.8	8.6	8.3	7.8
	5% WB	MCDBR	15.3	15.7	17.0	17.3	17.5	17.5	19.5	21.8	21.4	19.8	17.4	16.4
		MCWBR	7.8	7.8	7.7	7.8	8.8	9.4	10.2	9.7	9.9	8.9	8.2	7.9
(40) Clear-Sky Solar Irradiance	taub		0.394	0.394	0.387	0.367	0.352	0.361	0.364	0.423	0.528	0.457	0.410	0.401
	taud		2.432	2.426	2.442	2.486	2.486	2.452	2.427	2.219	1.933	2.205	2.348	2.394
	Ebn at Noon		301	297	291	283	274	264	267	260	246	275	295	300
	Edh at Noon		39	39	36	32	30	29	31	41	60	48	43	41
(44) All-Sky Solar Radiation	RadAvg		1846	1820	1712	1521	1197	1092	1179	1441	1588	1729	1905	1941
	RadStd		189	142	127	108	121	108	122	102	157	147	149	179

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (2 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page